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Research paper

Pharmacogenomic testing for major depressive disorder in British Columbia, Canada: Recommendations from a public deliberation☆

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ABSTRACT

Background: Pharmacogenomic (PGx) testing for major depressive disorder (MDD) can reduce healthcare spending and improve patient outcomes. However, concerns about health equity, data privacy, and secondary usage of data require deliberation. We sought recommendations from members of the general public for policymakers' consideration.
Methods: A four-day public deliberation was held in Vancouver, British Columbia (BC), from November–December 2024. Invitations were mailed to 15,000 randomly-selected households in BC. Participant selection maximized socio-demographic diversity. Participants received an information booklet and heard from expert speakers before discussing whether and how PGx testing for MDD should be implemented. Participants then generated and voted on recommendation statements. Finally, participants engaged with policy-/decision-maker panelists. Sessions were audio recorded and transcribed.
Results: Thirty participants generated 15 recommendations. Publicly funding PGx testing for MDD was supported under certain conditions. Participants unanimously called for strict test reporting standards, in line with evidence-based prescribing guidelines. Healthcare professional (HCP) education was seen as necessary, with ongoing access to PGx experts. To enhance accessibility and acceptability, the group suggested re-naming this "PGx testing (medication compatibility testing)" and conducting a targeted awareness/education campaign. However, participants wanted additional efficacy data collected across diverse ancestry groups to ensure PGx testing benefited everyone. While they agreed test results could be shared between HCPs, they wanted further public engagement on using PGx results in research without consent.
Conclusions: Knowledge gained throughout the event appeared to reduce concerns around data privacy, highlighting the importance of pre-implementation education and awareness. These recommendations provide guidance to policymakers considering PGx implementation.

1. Introduction

Major depressive disorder (MDD) is a common, recurrent condition that can seriously impact all aspects of one's life (World Health Organization, 2023). It is the single largest contributor to disability worldwide (Friedrich, 2017). There are many effective options for treating MDD, most commonly including using antidepressants alone or in combination with another type of therapy (Cuijpers et al., 2013; Lam et al., 2024). While many antidepressants are available, finding one that is efficacious and tolerable can be a lengthy process of trial-and-error. To illustrate, only 40–60% of patients respond to initial antidepressant therapy (Cipriani et al., 2018; Gkesoglou et al., 2022), up to 60%

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experience partial or no response to multiple trials of treatment (Rybak et al., 2021) (with each treatment trial lasting 4–6 weeks) (Lam et al., 2024), and 27% of people report adverse drug reactions (Mishra, 2013).

Variation in genes encoding the enzymes that metabolize medications explain up to 42% of the interindividual difference in medication response (Tansey et al., 2013). Pharmacogenomic (PGx) tests look for variation in three cytochrome P450 genes (*CYP2C19*, *CYP2D6*, *CYP2B6*) to predict the speed at which antidepressants are metabolized (Bousman et al., 2023). Systematic reviews demonstrate that PGx-guided antidepressant prescribing improves response and remission in adults with moderate-severe MDD (Arnone et al., 2023; Brown et al., 2022; Bunka et al., 2023; Santenna et al., 2024; Tesfamicael et al., 2024). However, the quality of the clinical trials is variable, resulting in high risk of bias and some heterogeneity in meta-analyses (Bunka et al., 2023). As a result, some groups, including the Canadian Network for Mood and Anxiety Treatments (CANMAT), have not recommended PGx testing for MDD (Lam et al., 2024). While there are fewer economic evaluations, a 2023 cost-effectiveness analysis predicted PGx implementation for MDD would save BC's health system nearly $1B CAD over 20 years (Ghanbarian et al., 2023). Similar conclusions of the economic benefit of PGx testing across psychiatric conditions have emerged in a large European cohort of the PREPARE study and a systematic review of economic evaluations (Karamperis et al., 2021; Skokou et al., 2024).

Despite evidence on effectiveness and cost-effectiveness, PGx testing for MDD is not publicly-available in Canada. There are several commercial laboratories (Maruf et al., 2020) offering testing and several health insurers that have policies that include PGx testing coverage (Brubaker-Plitt and Bousman, 2026; Genome BC, 2024). Recognizing the interest in this kind of testing back in 2015, Doctors of BC (the organization representing primary care physicians and committed to delivering high-quality care in BC) released a policy statement calling for federal regulation, reliability and validity standards, and public education (Doctors of BC, 2015a). Similar statements were recently echoed in a BC Ministry of Health PGx guidance document (BC Ministry of Health, 2023). Doctors of BC also listed preparatory recommendations for health system implementation, including a collaborative call for government, industry, physicians, and people with lived experience to develop genomic policymaking (Doctors of BC, 2015b). Taken together, PGx testing has been a topical consideration for some time in BC, and policy decisions may be imminent with mounting effectiveness and BC-specific cost-effectiveness evidence (Ghanbarian et al., 2023).

In single-payer healthcare systems like Canada, in which most urgent and direct healthcare costs are covered through public funding, it is important to consult broadly in decision-making, including the public. Major depression is common and will directly and/or indirectly (through friends and loved ones) impact almost everyone at some point, making public input into mental health care services critical. Any genetic test involves collecting and analyzing DNA. Therefore, PGx testing policies and regulations must balance the benefits of improved antidepressant prescribing with the social, ethical, and legal considerations around this type of testing, including the storage and access of this data.

While physicians, people with lived experience, and policymakers have contributed their input in studies about PGx testing for depression (Roberts et al., 2025; Slomp et al., 2023; Vest et al., 2025, 2020), we are only aware of one study that surveyed members of the public on design preferences for a PGx service (McDermott et al., 2024). This discrete choice experiment in the UK asked about public preferences for a small number of constrained options in a 15-minute survey, and focused on joint pain and low mood, rather than major depression (more enduring and severe than low mood) (American Psychiatric Association, 2024; World Health Organization, 2023). Beyond psychiatry, several studies have evaluated public preferences and perspectives on genomic sequencing (Etchegary et al., 2021), implementation of unspecified (no particular condition) PGx testing (Longo et al., 2016; Zhang et al., 2022), and PGx testing in oncology (Gornick et al., 2017). Incidental findings are not possible when analyzing the gene variants relevant to antidepressant metabolism, thus mitigating the ethical issues present with broader panel tests. Further, as major depression remains a stigmatized condition, there may be implementation issues that are specific to PGx testing in this context.

We, therefore, invited members of the British Columbian public to identify what is important to them about PGx testing for MDD, and to provide recommendations to policymakers in a public deliberation. Public deliberations provide an opportunity for members of the public to be actively involved in collective decision-making (A Sheedy, 2008). In consultation with policymakers, public input was sought on whether and how best to implement PGx testing for MDD in BC in an acceptable, equitable, and trustworthy manner. Here, we provide details of the public deliberation process and participant-generated recommendations for publicly-funding PGx testing for MDD in BC.

2. Methods

We followed well-established public deliberation methods (O'Doherty and Burgess, 2009) (see Box 1) to hold an in-person, four-day event in Vancouver, BC (November 16–17 & November 30–December 1, 2024). In preparation, the research team met several times over many months with clinicians, researchers, patient partners, methodologists, bioethicists, and policymakers to develop and refine four questions for deliberation:

1. *Should the BC health system fund PGx testing for adults with major depressive disorder*?
2. *How do we ensure equitable and inclusive access to PGx testing for adults with moderate to severe MDD in BC*?
3. *Under what circumstances should PGx test results be shared between healthcare professionals within the BC health system*?
4. *Should PGx test results for MDD be available for ethics-approved and authorized research, without asking individuals for consent*?

2.1. Recruitment

Recruitment was by postal invitation to 15,000 randomly selected households in all five regional health authorities in BC. The number of invitations sent per health authority was based on 2021 population estimates (Interior Health Data and Analytics Services, 2021). We oversampled in Northern and Interior Health regions, and in rural/remote areas, to ensure the perspectives of those living outside major metropolitan areas and those who self-identify as Indigenous were included. A direct mail company that partnered with Canada Post, printed and mailed the invitations in September 2024. The two-page invitation package included an invitation letter and flyer, which specified the topic, dates, reimbursement ($125 CAD/session; travel expenses; accommodation and meal per diems for out-of-town participants), and details about how to register interest. Sharing the invitation letter with others was encouraged.

Interested individuals completed an online recruitment survey with eligibility screening questions, socio-demographic items, and contact details for study notification. Expressions of interest were collected for 14 days. Adults (18+ years) living in BC, fluent in English, and who were able to attend all four days of the event were eligible. A target sample size of 30 participants was chosen to achieve diversity, while enabling thoughtful and considered discussion, in the constraints of a justifiable budget, and is similar to other public deliberations (O'Doherty and Burgess, 2009; Teng et al., 2019). Participant selection was guided by ensuring geographical representation (at least two individuals from each regional health authority (O'Doherty and Burgess, 2009); then, relatively more individuals from more populous regions), maximizing diversity on key socio-demographic variables of interest (gender, age group, self-reported ancestry, education, and urban/rural location), and including varying experience with MDD (direct, indirect, no known

Box 1
Key characteristics of public deliberations.

A public deliberation is a structured discussion in which members of the public deliberate about and make recommendations on imminent policy issues on which public values may conflict (Kim, 2019). Public deliberations aim to select a 'mini public' that represents the different needs, views, and values of those impacted by a policy decision (Goodin and Dryzek, 2006). This is done by selecting a small group of participants based on a range of life experiences, reflected in socio-demographic differences. Public deliberations highlight the potential social and ethical implications of a policy decision amongst the public, and so can enable better decision-making for policymakers (A Sheedy, 2008). They differ from focus groups or other forms of public engagement by the extent of information provision, the depth and length of discussions that occur over repeated interactions, the identification and weighing of potential trade-offs, and participant-generated recommendations that are voted upon and then shared with policymakers (Schwartz et al., 2021; Solomon and Abelson, 2012; Teng et al., 2019). Voting not only provides a gauge of the group's support for the recommendations, it also enables the opportunity to discuss and revise recommendations that are not well supported. Achieving consensus in the recommendations is desirable, but when this is not possible, points of persistent disagreement are noted. The outputs of public deliberations help broaden the base of relevant input into policy decisions, thus enhancing their legitimacy and public trustworthiness. They can be used to help resolve social dilemmas when scientific evidence is not sufficient to direct policy and there may be conflict in determining priorities for conflicting values (Rychetnik et al., 2013; Solomon and Abelson, 2012).

close contacts with MDD). We excluded health professionals, policymakers, lobbyists, and anyone who worked or volunteered for a mental health organization in an effort to avoid introducing hierarchy of opinions.

Selected potential participants were notified by email and asked to complete an online consent form. One reminder email was sent if a selected participant did not consent within a few days. After providing consent, participants were both emailed and mailed a welcome letter, event agenda, and an information booklet.

2.2. Providing background information

An information booklet (see Supplementary File 1) and presentations from expert speakers helped prepare participants for meaningful engagement as an informed public (Schwartz et al., 2021). The booklet was developed by the research team, underwent plain language checking, and was reviewed iteratively by experts, policymakers, and patient partners. The policy decision was clearly framed as offering PGx testing for adults with major depression after they had tried at least one antidepressant, rather than PGx testing compared to any other treatment options available for MDD. Participants were asked to read the booklet before the deliberation.

Six expert speakers were chosen to provide a range of perspectives on the deliberation topic; two people with lived/living experience of MDD (patient research partners on the research team), a family doctor with a special interest in mental health, a phamacogeneticist, a bioethicist, and an Indigenous knowledge keeper and hereditary Chief. It was important to have an Indigenous community leader inform this discussion because depression prevalence is high amongst many Indigenous communities (Hajizadeh et al., 2021; Hop Wo et al., 2020), harms have been committed against Indigenous Peoples in research (Bellamy and Hardy, 2015), and healthcare access and perspectives of the healthcare system vary from community-to-community (Adelson, 2005; Kim, 2019).

2.3. Structure of the event

All large group sessions were led by a trained facilitator with over 15 years of experience in public deliberations. Non-interacting observers (N = 5–10/day) were invited to attend the event, including policy-/decision-makers, researchers, clinicians, and people with lived experience. On Day 1, the six invited experts presented information about MDD, treatments for it, PGx testing, and relevant ethical and cultural considerations (see Box 2). A participant-driven question and answer session followed these presentations. Participants were then divided into four break-out groups of 6–8 participants. The small groups were formed to maximize diversity across the socio-demographic variables. The purpose of the small groups was to provide an opportunity for everyone to participate and to hear from a wide range of views. Small group discussions were supported by a trained facilitator (two trainees, two genetic counsellors) with prior knowledge of PGx for MDD, and the group nominated a speaker to report back to the large group with a summary of the discussion. In the first small group session, participants were asked to individually come up with at least three hopes and three concerns about offering PGx testing for MDD, which were written on Post-It Notes, placed on flip charts, and then discussed within the group. The group worked to produce a 'map' of interrelationships between the hopes (and concerns), and whether any hopes were in conflict with concerns.

Box 2
Format and purpose of daily public deliberation sessions.

Session	Focus	Purpose
Day 1	Expert speaker presentations (x6)	Provide balanced information Opportunity to ask questions Get to know other participants
Days 2–3	Discuss 4 deliberation questions Construct & vote on recommendations	Consider broad range of perspectives Explore different beliefs and reasons for them
Day 4	Policy and expert panel review & discussion Ratify & re-vote on recommendations	Enable policymakers to better understand rationale for recommendations Adds credibility to process and ensures participants' voices are heard Ensures knowledge gained is incorporated into final recommendations, votes

Over the remaining three days of the event, participants discussed and deliberated on the four deliberation questions. Each question was first introduced in the large group, then discussed in small group settings, so that participants could learn from other's views, values, and experiences in relation to the deliberation question. For question 4, the small groups were tasked with discussing four hypothetical research examples developed by the research team (see Supplementary File 2). The examples functioned as a contextual tool for this question and encouraged participants to consider trade-offs of sharing PGx test results for research purposes. After each small group, participants re-convened in the large group to deliberate and develop their recommendations. Participants' reasons for, against, or abstention for each recommendation were summarized on-screen by two research team members.

On the final day, five clinical experts and policymakers engaged with participants about the recommendations as part of a discussion panel. Afterwards, all recommendations were reviewed and refined (if necessary) by participants, and then re-voted on. This provided an opportunity for participants to incorporate what they had learned over the course of the event into the recommendations, including changing their mind on any of these statements. Subsequently, participants completed the Public and Patient Engagement Evaluation Tool (PPEET) (Abelson et al., 2016) and event satisfaction questions before departing.

2.4. Analysis of deliberative output

All sessions were audio-recorded and transcribed verbatim using Temi (www.temi.com). Transcriptions were checked and corrected against the audio recordings by the two note takers who attended the event. This paper reports the deliberative output (O'Doherty and Burgess, 2009), which consists of the recommendations, vote counts, and collective reasons for the votes. While the vote counts provide a quantitative measure of the degree of support for recommendations, the reasons underlying these are more significant, informative for policymakers, and can explain how close or far participants are from agreement (Moore and O'Doherty, 2014). This analysis is, therefore, focused on participants' reasons for voting, and less emphasis is given to interpreting numerical differences in votes. A qualitative analysis of implementation considerations pertaining to these recommendations is underway and will be reported separately.

2.5. Ethics

This study was approved by the University of British Columbia Behavioural Research Ethics Board (#H24-01354) and Vancouver Island Health Authority Research Ethics & Compliance Office. All participants provided informed consent online prior to participation.

3. Results

Of the 403 expressions of interest received, 284 people were potentially eligible, and 45 were invited to participate. Thirty-one participants provided consent, but one subsequently declined. The remaining 30 participants came from all five regional health authorities, including 17% (5/30) from rural/remote locations (Table 1). Retention of participants across all four days was 93% (28/30): 30 took part in Days 1–2, 29 in Day 3, and 28 in Day 4. Participant withdrawal was for medical reasons, unrelated to the event.

Participants collectively developed 15 recommendations (see Table 2), grouped into five categories: 1) Support for PGx testing; 2) Requisite conditions for implementation; 3) Enhancing accessibility, utility, and equity; 4) Test result sharing; and 5) Secondary (non-clinical) uses of data. There was also one persistent point of disagreement relating to sharing data for secondary research, in which participants were split in terms of being in favour of, against, or abstaining from the suggested course of action (no clear majority).

Table 1
Self-reported socio-demographic characteristics of deliberation participants ($N = 30$).

	N (%)
Gender	
Woman	17 (57)
Man	13 (43)
Ancestry[a]	
Black	3 (10)
East Asian	5 (17)
European	13 (43)
Indigenous	5 (17)
South Asian	2 (7)
Southeast Asian	2 (7)
Regional Health Authority	
Fraser Health	10 (33)
Interior Health	4 (13)
Northern Health	4 (13)
Vancouver Island Health Authority	6 (20)
Vancouver Coastal Health	6 (20)
Country of birth	
Canada	23 (77)
Other	7 (23)
Age group (years)	
18–24	4 (13)
25–34	5 (17)
35–49	7 (23)
50–64	8 (27)
65+	6 (20)
Highest level of education attained	
High school diploma/certificate	6 (20)
College/apprenticeship (non-university)	9 (30)
Some university	1 (3)
University degree or diploma (BA/BSc level)	13 (43)
Professional or graduate degree	1 (3)
Main activity	
Working at a paid job/business	16 (53)
Retired	8 (27)
Looking for paid work	2 (7)
Going to school	1 (3)
Household work	1 (3)
Caring for children	1 (3)
Long-term illness	1 (3)
Chronic condition (personal or dependent)	
Yes	14 (47)
No	16 (53)
Experience with major depressive disorder (MDD)	
Diagnosed with MDD	6 (20)
No MDD, but close others have MDD	17 (57)
No MDD, no one close has MDD	7 (23)
Ever had any kind of PGx testing	
Yes	1 (3)
No	29 (97)
Income ($ CAD)	
Less than $20K	5 (17)
$20K-$34,999	3 (10)
$35K-$49,999	4 (13)
$50K-$79,999	8 (27)
$80K-$99,999	5 (17)
$100K+	5 (17)
Religion	
No religion	16 (53)
Christian (United, Baptist, Anglican, Catholic)[b]	9 (30)
Buddhist	1 (3)
Aboriginal spirituality	1 (3)
Other	3 (10)

NB: percentages may not always sum to 100% due to rounding.

[a] Categories come from guidance on race-based and Indigenous data collection standards provided by the Canadian Institute for Health Information (see Table 2 Race-based question and responses) (Canadian Institute for Health Information, 2022).

[b] Denominations of Christianity were asked separately, but have been grouped here for ease of presentation. Religious categories were based on the Canadian Census categories.

Table 2
Recommendations and distribution of votes.

Recommendations	Yes	No	Abstain
Support for PGx testing			
1. The BC health system should fund PGx testing for adults diagnosed with MDD under certain conditions.	22	1	5
Requisite conditions for implementation			
2. Should PGx testing be funded by the BC health system, tests must meet consistently strict quality standards for reporting results.	28	0	0
3. When reporting PGx results, labs must follow evidence-based prescribing guidelines.	28	0	0
4. There should be education for healthcare professionals to be aware of PGx tests for MDD and be prepared to interpret them to guide prescriptions.	28	0	0
5. Prescribers should have access to clinical experts in PGx for MDD to support prescribing decisions (ex: RACE).	28	0	0
6. Current policy of destroying biological samples is sufficient; no additional security and/or legal measures are required.	28	0	0
7. No additional security and/or legal measures to protect genetic information are required, beyond what is already in place to protect genetic information.	20	5	3
8. No additional security and/or legal measures to protect patient confidentiality are required for test results about metabolizer status in the medical record.	25	3	0
Enhancing accessibility, utility, & equity			
9. PGx testing for MDD should be called “PGx testing (medication compatibility testing)” or something similar.	25	1	2
10. If PGx testing is to move forward, there must be a targeted awareness/education campaign about the possibility of PGx testing as one of the options for depression care (care for MDD).	18	3	7
11. Efforts should be made to collect data across diverse ancestry groups to ensure broad applicability of PGx testing for MDD.	26	1	1
Test result sharing			
12. PGx test results for MDD may be shared between healthcare professionals directly involved in the patient's care.	28	0	0
13. PGx test results for MDD should follow the same protocols for consent and sharing test results with healthcare professionals as other sensitive test results.	27	1	0
Secondary (non-clinical) uses of data			
14. Any results of PGx research using public data done by private companies needs to be published or put into public domain.	25	0	3
15. Further public engagement on consent for secondary usage of data with PGx results needs to be conducted.	18	2	8

NB: total number of votes sum to 28 because 2 participants were absent on Day 4 of the event.
BC = British Columbia, Canada; PGx = pharmacogenomic; MDD = major depressive disorder; RACE = Rapid Access to Consultative Expertise (i.e., a telephone support line for healthcare professionals).

1. Support for PGx testing

Most participants were in favour of the health system funding PGx testing for adults diagnosed with MDD (Recommendation 1). Participants understood that PGx testing could be a useful tool. Although participants acknowledged gaps in the evidence base, most felt comfortable for the science to improve post-implementation. The wide-ranging detrimental impacts of depression on people's lives were key focal points in the discussion, and this drove participants' desire to improve mental health care. Some participants did specify that having a diagnosis of MDD was important, to ensure that appropriate steps in depression care had been taken. However, it was the inclusion of the ‘diagnosis’ qualifier – the implication that certain tests would have to precede PGx testing and by accessing specific HCPs (when this is not always possible or preferred) – that led some to abstain. Others abstained because they wanted efficacy evidence that included more diverse ancestry groups, to ensure equal benefit for people of non-European ancestries. Some participants called for funding only in certain situations, such as for low income groups.

2. Requisite conditions for implementation

Within Recommendation 1, participants stated that they were in favour of publicly funding PGx testing under certain conditions. These conditions comprised seven subsequent recommendations, which fell into three categories: 1) test reporting standards; 2) HCP education and ongoing support; and 3) protecting biological samples, genetic information, and test report results. There was strong support for all of these recommendations, with only protection of genetic information and PGx test results (Recommendations 7–8) lacking unanimous support.

For the test reporting standards conditions, participants were concerned that different laboratories might end up with variations in the PGx test reports, and so called for adherence to ‘consistently strict reporting standards’ (Recommendation 2). This was about test report uniformity through standard application of rules. All participants wanted the medication-specific recommendations in the PGx test reports to follow evidence-based guidelines (Recommendation 3). In this case, participants wanted transparency, as well as the assurance that prescribing recommendations were based on reputable evidence. For some, this was grounded in the assertion that laboratories should only report results, rather than make interpretations or recommendations. Although participants did not specify which expert guidelines should be used, Clinical Pharmacogenetics Implementation Consortium (CPIC) was referenced as an example, noting that these are updated regularly.

Second, everyone voted in favour of educating HCPs about PGx tests for MDD and how to interpret test results (Recommendation 4), as well as the provision of ongoing PGx expert support and guidance (Recommendation 5). Although many participants imagined family doctors as the primary provider of PGx-based care, participants wanted to be non-specific in Recommendation 4, to allow for broader inclusion of HCPs, including specialists, nurse practitioners, and pharmacists. Both benefits and concerns were emphasized about pharmacists prescribing antidepressants, however. Given the complexity of depression, prescribing decisions, and the evolving evidence base around PGx testing, participants also highlighted the importance of HCPs being able to consult with a clinical PGx expert. Some were aware of existing phone and email/app support systems in place for HCPs in the province, and suggested these could be viable solutions for expansion to PGx-based guidance. Participants agreed that a medical or genetics expert was required to corroborate prescribing decisions and answer PGx questions.

The third category of conditions asserted that no additional security and/or legal measures were required to protect biological test samples, genetic information, and the PGx test results, beyond the current practices and policies. In early discussions, participants' concerns varied between these three types of data, and so they were broken out into separate statements by the research team to parse out the specific issues (Recommendations 6–8). All participants agreed that biological samples (e.g., blood, saliva) should be destroyed after analysis (Recommendation 6). Next, most participants believed the current measures in place to protect genetic information were adequate (Recommendation 7), given genetic testing already occurs within the BC health system. One participant noted that the desire for more data security might never end, which seemed to attenuate calls for more protections. Reasons for abstention or disagreement centred on concerns about insurance companies using the data to deny coverage, how much protection the Genetic Non-Discrimination Act (GNDA) afforded, and the desire for continual re-evaluation and updates to these measures (in anticipation of substantial change within the next few years). There was, however, strong agreement that PGx test results for MDD could reside in a patient's electronic medical record (EMR) without the need for additional protections (Recommendation 8). Participants implied it would be more serious for a data breach to their full EMR record than just the PGx test results, that those overly concerned about a data breach could just opt out of PGx testing, and that knowing these test results was no worse than someone knowing of a depression diagnosis (which already occurs when filling an antidepressant prescription). Objections to the recommendation included concern about whether additional medical information (e.g., other medications) could be deduced from the PGx test results, a desire to protect all personal information and prevent any discriminatory acts based on, as well as enact penalties for improper use of, PGx test results.

3. Enhancing accessibility, utility, and equity

To enhance accessibility, participants first suggested broadening the test name to "PGx testing (medication compatibility testing)" (Recommendation 9). Most indicated there was a 'scary' connotation with genetic testing, and felt that this labelling would be received more positively. Some noted the bracketed term might be easily understood by lay people, yet they did not want to conceal the genetic testing component. Reasons for abstention or disagreement were that policy-makers should be tasked with naming decisions, re-branding could foster distrust (and transparency was paramount), and the belief that people would be able to comprehend 'PGx testing'.

A second accessibility-related recommendation was to conduct a targeted awareness/education campaign (Recommendation 10). Since PGx testing would be a new component of depression care, participants wanted to ensure that it was widely known, so that patients could request it. Others raised the point that many people have had negative experiences with antidepressants, and the awareness that this tool could improve future prescribing was important to communicate. Given mental health stigma, some thought the campaign needs to reach adolescents/youth; for others, the campaign was about educating HCPs about how to use PGx test information, as well as involving community and outreach workers (may have more contact with rural/remote or harder-to-reach populations). Some imagined that trusted ambassadors, such as well-known athletes or Indigenous Elders, could deliver tailored awareness messages to specific population groups and may be more influential than HCPs. While the majority of participants voted in favour of this recommendation, several participants abstained or disagreed over word choice. For example, one participant said "must" was too definite and they didn't want education mandated, while another wanted "when PGx testing…" instead of "if". Some argued that a 'campaign' could make too big an issue over this testing, awareness of depression care (not PGx testing) was needed, only HCPs needed more education/information, or that it was not clear who the campaign was targeting.

Finally, nearly all participants wanted more efficacy data collected to ensure PGx testing benefited diverse ancestry groups (Recommendation 11). This was about increasing representation of non-European groups, who have typically not been included in randomized controlled trials. One person agreed with the sentiment of this statement, but abstained because they did not feel this was policy-relevant. There was some concern about racially-targeted data collection, and confusion about why race was relevant for this type of testing.

4. Test result sharing

Participants unanimously supported sharing PGx test results for MDD between HCPs directly involved in a patient's care (Recommendation 12). They said that the test information "wasn't a big deal" and it didn't show as much information as they originally thought, with some likening it to knowing an individual's blood type. Participants wanted this information accessible across regional health authorities to enhance communication with HCPs wherever healthcare is provided, and trusted the health system to protect their data. Some suggested that anyone prescribing medication should be able to see the test result, even if they were not treating depression for the patient. Some mentioned that the more information that doctors and specialists have, within a holistic care context, the better they can discern proper treatment/medication for the patient. Many expressed a desire for these test results to be available to patients through existing online patient health information portals.

In terms of protocols around consent and sharing test results, all but one participant agreed that these should align with other procedures in place for sensitive tests (Recommendation 13). Both HIV and other publicly-funded PGx tests in the province (e.g., *HLA-B*58:01* testing for allopurinol) were cited as examples. Those in favour articulated that results sharing between HCPs should be explained to patients at the time of testing, with appropriate steps taken to account for language (e.g., translations) and cultural sensitivities. The participant who objected wanted test results widely shared to provide the most patient benefit, and was concerned about increasing mental health stigma if the PGx test was classed as a sensitive test result.

5. Secondary (non-clinical) uses of data

The hypothetical research examples (Supplementary File 2) provided in the small groups informed large group discussions and recommendations on using PGx test results in future research. Research that provided health or service delivery benefits was deemed important by participants – the potential benefits appeared to outweigh the risk of a potential data breach – but there were mixed feelings when it came to research by pharmaceutical or for-profit companies. Recognizing that research is predominantly conducted 'for the greater social good', but to ensure accountability and public benefit, participants recommended that any PGx research using public data that is commissioned by private companies must be published or made publicly available (Recommendation 14). No one voted against this recommendation, but reasons for abstention were that publication involves effort and costs and does not guarantee the information is accessed, consent was deemed necessary before publication, and the current policy on research using health data was unclear.

Research was seen as important and beneficial, but whether prior consent was required to use PGx test results was "a big and divisive topic," as described by one participant. It is worth noting that the issue with consent was firmly centered on secondary use of the results; the group agreed that consent was implied and not required when seeking PGx testing for clinical care. This discussion resulted in a persistent point of disagreement, in which there was a three-way split in votes for, against, and abstentions to research without prior consent. Some participants held a restrictive position, explicitly requiring either i) informed consent in advance of PGx testing or before each study, regardless of the cost or time implications of re-identification, or ii) an opt-out clause. Others specified that certain conditions needed to be met to enable agreement with future research without consent, such as the data must remain in BC or Canada, and pharmaceutical companies must abide by the Five SAFES framework (Desai et al., 2016) for accessing sensitive data. The remaining position supported research and did not think consent or any other new conditions needed to be introduced. Regardless of the position taken, most participants were surprised at learning that their health service use data was already being included in authorized research without their consent. Many did not feel they had enough time to fully understand the processes, procedures, protections, and other issues for them to develop clear, collective recommendations. A resolution came in the form of Recommendation 15: further public engagement is needed on the topic of consent for secondary use of PGx test results. Although the majority voted in favour of this recommendation, there were several abstentions; namely, people wanted to move forward with the positions they had already articulated (e.g., requiring informed consent, trusting the existing protections in place) and some felt that mandating public consultation was too strong.

4. Discussion

This was the first pubic deliberation about providing PGx testing as part of routine depression care. All participants recognized the need for improved mental health care, and PGx testing was seen as one way to address this. Participants voted in favour of offering PGx testing for MDD and sharing results between HCPs in the circle of care, but specified certain implementation-related conditions; namely, standardized laboratory procedures and test reporting, adequate education and ongoing support for HCPs, and continuation of current practices and protocols around protecting biological samples, genetic information, and PGx test results. Collectively, the participant-generated recommendations can inform policymaking around the provision of acceptable and trustworthy implementation of PGx testing for MDD, and offer important lessons for jurisdictions considering how best to implement PGx testing.

A core feature of public deliberations is for participants to exchange ideas, experiences, and proposals over the course of their conversations. This means that participants may change their positions as they learn and are exposed to other perspectives, before the recommendations of the group are finalized. It is beyond the scope of the paper to to examine in detail the dynamics that preceded the finalization of all recommendations, but we discuss the trajectories whereby selected recommendations were finalized. To begin with, many participants in this public deliberation said that they came to this event with ethical, privacy, and security concerns about PGx testing for MDD precisely because this is a genetic test. This partly drove the suggestion to re-name this test to counter some of the negative genetic testing connotations (Recommendation 9), as well as increase test accessibility. These same concerns underpinned Recommendations 6–8, which originally called for additional security and/or legal measures for biological samples, genetic information, and PGx test results. Several participants had raised privacy and security concerns for these three different types of test-related data on Day 2, resulting in a split between those in favour and those against greater protections of genetic information and PGx test results storage. These concerns were not unexpected, given the impetus for this public deliberation was borne out of findings from prior studies (Haddy et al., 2010) and personal communications (Dr. Jesse Swen, virtual meeting, 04/04/2023) that some patients, members of the public, and HCPs are wary of the storage of genetic test results in the EMR. As has been found for other genomic issues in previous deliberations (Gornick et al., 2017), however, the understanding that was gained through the deliberative process appeared to cause many people to shift their perspective. Over time, several participants came to view the results of this test as ‘benign’ – akin to knowing one's blood type – and lacking identifying or stigmatizing information. When the recommendations were ratified and re-voted on at the end of the event, the majority of participants no longer felt the need for additional precautions or measures.

The attenuation of privacy and security concerns has important implications for PGx pre-implementation. First, it denotes the importance of providing in-depth education about PGx testing for MDD (i.e., Recommendation 10), which may be a crucial precursor to comfort with and uptake of this testing. This recommendation is similar to findings from other studies on genomic testing. For instance, participants similarly emphasized the importance of a public awareness campaign about genomic sequencing results before any clinical consultation for this testing (Etchegary et al., 2021). While some existing PGx programs provide information online or in videos (Mayo Clinic, 2017), others employ pharmacists or genetic counsellors to conduct PGx pre-consultation appointments (Hoffecker et al., 2024). Second, the nature of discussions in this public deliberation may offer clues about what needs to be addressed in education campaigns to lessen such concerns. For instance, careful explanation of the limited number of specific genes that PGx tests for MDD analyze (versus whole genome sequencing), what happens to the sample after collection through to storing and reporting results, and what type of information is and isn't provided in a PGx test report. Similar to previous studies (Haddy et al., 2010), however, a few people remained cautious and wanted greater protection against insurance companies, in particular, from obtaining or selling information related to genetic testing. Although participants in this public deliberation read about Canada's Genetic Non-Discrimination Act in the information booklet, they wanted stronger stipulations, as well as penalties for discriminatory acts based on genetic testing. These are reasonable suggestions that, if addressed, could promote greater public acceptance with PGx testing.

Despite the stigma that still exists around having a mental health condition, all participants in this study agreed that PGx test results for MDD should be shared with any healthcare professionals directly involved in the patient's care (Recommendations 12–13). Currently, laboratory test results in BC are only shared with the ordering physician and any other clinically-relevant physician listed on the test requisition. While most participants mentioned shared access amongst those involved in mental health care, several participants also mentioned advantages if pharmacists, hospital emergency physicians, and specialists outside of mental health also had access. For some, it was as simple as anyone prescribing medication should have this information. The group drew on personal experience and also imagined several different scenarios in which it would be useful for a range of HCPs across regional boundaries and sectors of care to have access to these results, such as when travelling, re-locating, and for unattached patients without a dedicated family physician. Canadian participants in recent town hall discussions (Etchegary et al., 2021) and a large UK survey (McDermott et al., 2024) were also in favour of sharing genomic sequencing and PGx test results with other healthcare professionals, and a US survey revealed most participants trusted HCPs to keep PGx data private and were willing to contribute to a statewide PGx database (Zhang et al., 2022). To this end, some of the current public deliberation participants highlighted that accessibility and security would be enhanced if the results were stored electronically, such as within the EMR. Similar to recent PGx service design preferences with members of the public in the UK (McDermott et al., 2024), several participants in the current public deliberation also mentioned wanting to ensure patients had electronic access to these results. At present, no genetic test results are electronically available to patients in BC. Policymakers may need to reconsider whether this is perpetuating genetic exceptionalism, given most participants did not view PGx test results as sensitive, personally identifying, or different from other kinds of laboratory tests by the end of the deliberation. Interestingly, some believed that an added benefit of having a genetic test available was that it could reduce stigma around mental health conditions, making it easier for individuals to raise mental health issues and unsuccessful treatment trials with their HCPs, especially amongst those who might otherwise not seek treatment from HCPs.

This study provided the opportunity for British Columbians to contribute to policy decision-making in mental health care. This was a group of informed members of the public who came together to share their values and perspectives on publicly-funding PGx testing. Participants respectfully tried to understand each other's points of view, reflected back what others had said, and acknowledged the importance of perspectives that might differ from their own. They were highly engaged and motivated throughout the four full days of this event. This was marked by frequent referencing to information sources (booklet, expert speakers), reviewing and sharing back the reference materials they had read in their own time, expressing the desire to continue discussions overtime, and the high retention rate across two weekends (93%).

It is important to emphasize that the goals of a public deliberation are not to produce knowledge that is generalizable to the wider population. Rather, the goal is to open up policy discussion to a wide range of carefully considered perspectives from members of the public. This particular group came together to deliberate on this specific topic at a particular point in time, and their discussions focused on both current events and personal experiences. However, this was a sociodemographically diverse group, some with and without direct experience of depression, which is likely to reflect diversity in their views and experiences. We also acknowledge that discussion was more limited on the fourth deliberation question about using PGx test results for research without explicit consent, than the previous three questions. Participants indicated that they did not have enough time to understand and discuss all of the issues that they wanted to about PGx testing for MDD in this regard, which resulted in a persistent point of disagreement.

5. Conclusions

PGx testing involves ethical, privacy, and security considerations, but members of the public saw it as a worthwhile tool that should be publicly-available through the health system to improve the treatment of major depression. Concerns about genetic testing decreased through the knowledge gained in this public deliberation, underscoring the importance of providing education and raising awareness prior to implementation. PGx testing for MDD was seen as similar to other laboratory tests, but members of the public wanted results shared between healthcare professionals. Efforts are needed to enhance accessibility and ensure PGx testing benefits members of diverse ancestry groups. To enable acceptable and trustworthy implementation, policies need to ensure standardized laboratory procedures and PGx test reporting, education and ongoing support for HCPs, and upholding current protocols around protecting biological samples, genetic information, and test results.

Abbreviations

CANMAT Canadian Network for Mood and Anxiety Treatments
CPIC Clinical Pharmacogenetics Implementation Consortium
HCP healthcare professional
MDD major depressive disorder
PGx testing pharmacogenomic testing
RACE Rapid Access to Consultative Expertise

CRediT authorship contribution statement

Louisa Edwards: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Formal analysis, Conceptualization. **Colene Bentley:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Kieran C. O'Doherty:** Writing – review & editing, Methodology, Conceptualization. **Mary Bunka:** Writing – review & editing, Methodology, Investigation, Data curation. **Carly Pistawka:** Writing – review & editing, Methodology. **Linda Riches:** Writing – review & editing, Methodology. **Lisa Ridgway:** Writing – review & editing, Methodology. **Chad Bousman:** Writing – review & editing, Methodology. **Jehannine Austin:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization. **Stirling Bryan:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jad.2026.121910.

Data availability

The data used during the current study are not publicly available in order to protect the confidentiality of participants. Deidentified data may be available from the corresponding author on reasonable request.

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